

CSE 340 Quiz-4 – Section 29

Full Marks: 20

Date: Jan 3, 2026

Time: 30 minutes

Name: _____

Roll: _____

1. Find the cache index, Tag, and offset bits for the memory address = 297, given that the Block size is 32 bits and the cache size is 64 bytes. [5 marks]

Block Size = 32 bits = 4 Bytes → So offset = 2 bits

Cache Size = 64 Bytes = 64/4 blocks = 16 blocks → So index = 4 bits

Memory address = 297 = 100 1010 01
 Tag Index Offset

2. Cache table:

Index	Valid bit	Tag	Data(32 bits)
000	1	100	Data
001	0		
010	1	011	Data
011	1	001	Data
100	0		
101	1	000	Data
110	1	010	Data
111	0		

1. 10010001
2. 11101010
3. 01011011
4. 10010011
5. 11111010

[8+2+3+2 marks]

- a) For the given memory access, determine if it will be a hit or a miss.
- b) Calculate the hit and miss ratio.
- c) Find the AMAT if the hit time is 2s and the miss penalty is 5s.
- d) If the CPU has a 2s clock, then determine how many cycles are required per instruction on average.

Block Size = 32 bits = 4 Bytes → So offset = 2 bits

Cache Size = 8 blocks → So index = 3 bits

	Tag	Index	Offset	
1.	100	100	01	→ Miss
2.	111	010	10	→ Miss
3.	010	110	11	→ Hit
4.	100	100	11	→ Hit
5.	111	110	10	→ Miss

So hit ratio = $\frac{2}{5}$, miss ratio = $\frac{3}{5}$

AMAT = hit time + miss ratio*miss penalty = $2 + \frac{3}{5} * 5 = 5$ s

Clock period = 2s

Clocks per instruction = AMAT/Clock period = $5/2 = 2.5$